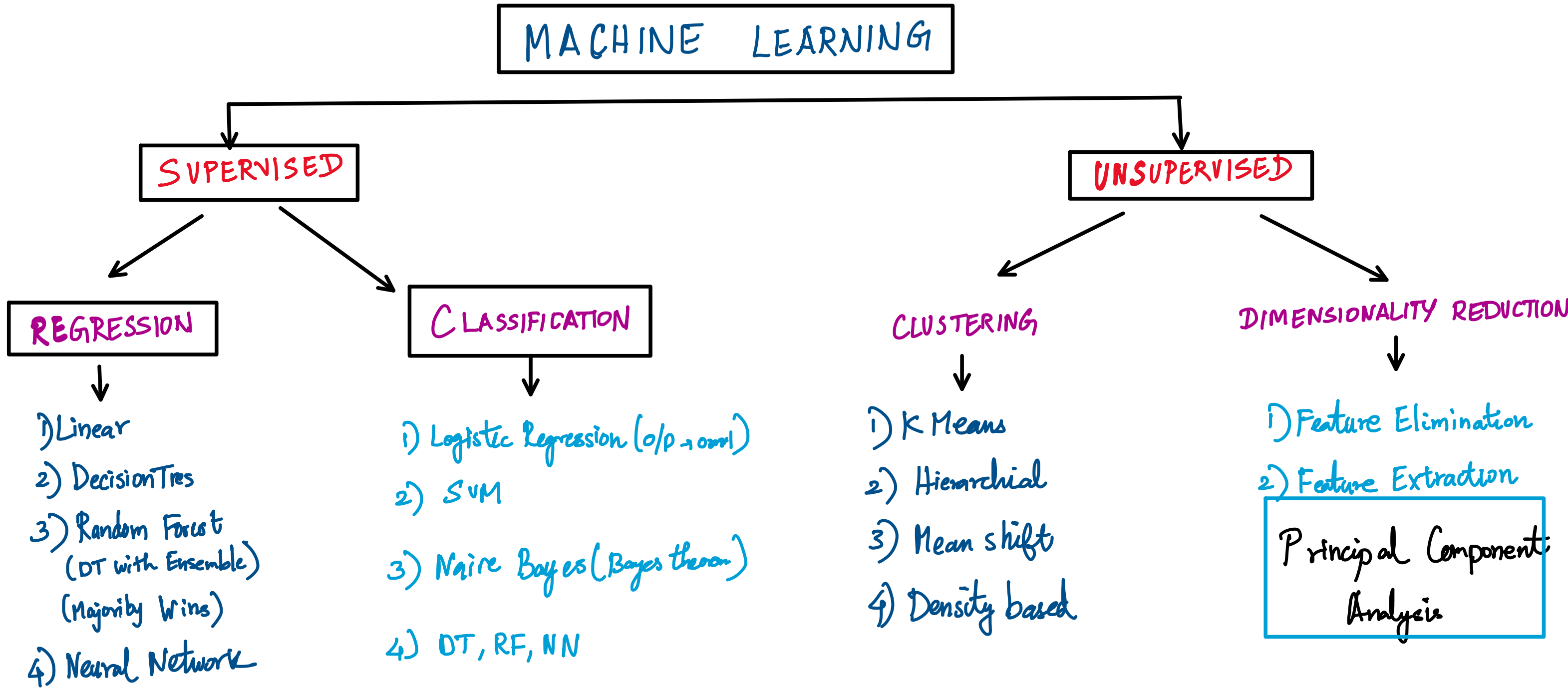


MACHINE LEARNING

Monday, June 30, 2025 2:24 PM



ML & NN MODELS PREFERRED IN MANUFACTURING APPLICATIONS

ML MODELS

ML MODELS PROF. SUGGESTED

- 1) LR
2) Ridge / Lasso
3) Logistic Regression
4) MARS
5) Decision tree
6) MLR

- Supervised Learning
- Linear Models

Linear Regression / Ridge / Lasso

Logistic Regression

Tree-Based Models

Decision Trees

Random Forest / Extra Trees

Gradient Boosted Trees (XGBoost, LightGBM, CatBoost)

Support Vector Machines (SVC / SVR / One-Class)

Instance-Based Models (KNN)

Neural Networks

MLP / Feedforward

CNN / Vision Models

RNN / LSTM / GRU / Transformers

Autoencoders (for anomaly)

Bayesian Models

Naive Bayes

Bayesian Regression

Ensembles

Bagging

Boosting

Stacking

NN MODELS

Data Types

- 1) Structured → Predict Quality, energy usage
2) Time series → Predict Maintenance
3) Image Data → Visual Defects

NN Models used in JMP → ≥ 300 Records

- 1) FNN or Shallow → < 10 Features
2) Feature Dimensionality → > 30 Features

⊗ If it has high Cardinality → Encoding / Grouping
Multi Colinearity → Remove / Combine Using PCA

Model	Task	Data Used	Performance
Structured TDNN/Autoencoder	Anomaly detection	~150 discrete/process sensors/events	+64% better misclassification reduction
DeepESN	Anomaly & maintenance forecasting	88 energy features	Faster fault detection than LSTM
LSTM	Data recovery	Strain/stress sensors on civil structures	MSE=0.6, MAPE<15%
GNN	RUL estimation	Engine sensor time-series	Modular interpretability via attention
TQRNN	Failure prediction	Industrial machine sensor streams	+11% yield; 70.8% accuracy

MANUFACTURING APPLICATIONS

ANOMALY DETECTIONS

Model	Type	Best For	Data Needed
Isolation Forest	Ensemble	Tabular, sensor logs	Structured
One-Class SVM	Kernel-Based	Clean normal data	Small, labeled
Autoencoder	Neural Net	Complex high-dim data	Tabular or time-series
LSTM Autoencoder	Neural Net	Time-series anomaly detection	Sequential sensor
DBSCAN	Clustering	Spatial/clustered sensor data	Moderate
LOF	Density-Based	Process variation anomalies	Structured
LSTM	Recurrent NN	Predictive maintenance, trends	Time-series

Application	Model Used
Motor failure prediction	LSTM, Isolation Forest
Abnormal cycle time	Autoencoder, DBSCAN
Sudden power surges	LOF, Isolation Forest
Tool wear anomaly	One-Class SVM, LSTM Autoencoder
Visual defect detection	CNNs + Anomaly Scoring (advanced vision)

OTHER APPLICATIONS

Application	Models Used	Dataset Size (Guideline)	Key Features	Tips to Avoid Bias / Overfitting
Anomaly Detection	Isolation Forest, Autoencoder, LSTM, LOF	500+ rows × 5–30 features (min)	Sensor readings, timestamps, event type	Normalize, avoid duplicate time points, balance data
Predictive Maintenance	LSTM, Random Forest, DeepESN, XGBoost	1,000+ time points per asset	Vibration, current, temperature, runtime counters	Use time lags; stratify by failure class
Defect Detection	CNN, T-CNN, Autoencoder, SVM	1,000+ labeled images or 500+ structured logs	Image pixels, part ID, surface features	Augment minority classes; avoid duplicate parts
Quality Prediction	MLR, MARS, ANN, Gradient Boosted Trees	300+ rows × 10–50 features	Temperature, pressure, resin type, cycle time	Remove correlated features; validate on new batches
Yield Optimization	Random Forest, XGBoost, SVM, ANN	1,000+ samples across time or batches	Process settings, material grade, humidity	Use cross-validation; avoid overfitting to one batch
Energy Consumption Modeling	MLR, XGBoost, LSTM, DeepESN	1,000+ time steps × 10–20 sensors	Load, RPM, voltage, shift ID	Account for daily/weekly cycles
Cycle Time Estimation	Decision Tree, XGBoost, MLP, k-NN	500–5,000 rows	Machine type, job type, speed, part complexity	Remove outliers; bin job types if sparse
RUL (Remaining Useful Life)	LSTM, RNN, GNN, Kalman Filters	1,000+ time points per machine/component	Sensor sequences, timestamps, cumulative usage	Segment properly; pad/truncate sequences uniformly

Use Case	Model Type	Industry	Dataset Description
Predict machine tool wear	Bayesian ANN, RF	Aerospace	Time-series sensor logs (force, torque)
Spot visual defects on circuit boards	CNN, Autoencoder	Electronics	10,000+ labeled PCB images
Predict concrete strength	MLR, PINN	Construction	Tabular mix data (temp, admixture ratios)
Estimate remaining battery life	LSTM, GNN	Automotive	5,000 cycles of voltage/temp readings
Forecast yield from process settings	XGBoost, SVM	Semiconductor	20+ process variables over 1,000 wafers
Detect abnormal power usage	Isolation Forest	Smart Manufacturing	88 energy signals per line, per hour

Best Model Type per Manufacturing Use Case

Task	Recommended Model(s)
Quality prediction	Random Forest, XGBoost, MLP
Visual defect detection	CNN, T-CNN
Time-based failure prediction	LSTM, DeepESN, Transformer
Anomaly detection	Autoencoder, Isolation Forest
Mixed categorical/numerical	CatBoost, LightGBM
Root cause analysis	Decision Trees, SHAP on GBDT

COMPLETE ML/NN HIERARCHY

1. Supervised Learning

- A. Regression Models

Linear Regression

Ridge Regression

Lasso Regression

ElasticNet Regression

Polynomial Regression

Support Vector Regression (SVR)

Decision Tree Regression

Random Forest Regression

Gradient Boosting Regression (GBR)

XGBoost Regression

LightGBM Regression

CatBoost Regression

KNN Regression

MARS (Multivariate Adaptive Regression Splines)

Neural Networks (for Regression)

Bayesian Regression

B. Classification Models

Logistic Regression

Support Vector Machines (SVM)

K-Nearest Neighbors (KNN)

Naive Bayes (Gaussian, Multinomial, Bernoulli)

Decision Trees

Random Forest

Gradient Boosted Trees (XGBoost, LightGBM, CatBoost)

Neural Networks / Deep Learning (MLP, CNNs)

Quadratic Discriminant Analysis (QDA)

Linear Discriminant Analysis (LDA)

Extra Trees Classifier

Ensemble Methods

Bagging

Boosting

Stacking

Voting Classifiers

2. Unsupervised Learning

- A. Clustering Models

K-Means

Hierarchical Clustering

Mean Shift

DBSCAN

OPTICS

Gaussian Mixture Models (GMM)

Spectral Clustering

Affinity Propagation

Agglomerative Clustering

Birch Clustering

Self-Organizing Maps (SOMs)

B. Dimensionality Reduction

Principal Component Analysis (PCA)

Independent Component Analysis (ICA)

t-SNE

UMAP

Factor Analysis

Autoencoders

Locally Linear Embedding (LLE)

ISOMAP

Feature Selection Techniques

Variance Threshold

Recursive Feature Elimination (RFE)

3. Semi-Supervised Learning

- Self-Training

Label Propagation

Label Spreading

Semi-Supervised SVM

Co-Training

Graph-Based Models

4. Reinforcement Learning

Q-Learning

Deep Q-Network (DQN)

SARSA

Policy Gradient Methods

REINFORCE Algorithm

Actor-Critic Methods

Proximal Policy Optimization (PPO)

Twin Delayed DDPG (TD3)

Soft Actor-Critic (SAC)

Multi-Armed Bandits

Common ML Use Cases in Manufacturing

- Predictive Maintenance & Yield Prediction

Regression Trees

MARS

Random Forests

Neural Networks

Fault Detection / Anomaly Patterns

Clustering (K-Means, DBSCAN)

Sensor Data Compression

PCA

Autoencoders

Classification & Anomaly Detection

XGBoost

LightGBM

Robotics & Real-Time Control

Q-Learning